

Research progress in hydrofoil cavitation prediction and suppression methods

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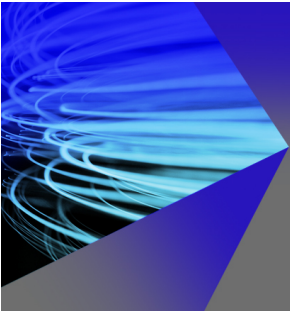
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ABSTRACT

To reduce the adverse damage caused by cavitation phenomena to the hydraulic machinery, such as surface erosion of the equipment, increased mechanical vibration, and decreased service life, this review summarizes from the aspects of cavitation instability mechanisms, cavitation prediction methods, and cavitation suppression methods. In terms of cavitation flow instability mechanisms, two main mechanisms that affect the shedding of cloud cavitation, reentrant jet, and bubbly shock wave, were thoroughly summarized. It is pointed out that the shedding behavior of the cavity is greatly influenced by the thickness of the reentrant jet relative to the cavity, and the bubbly shock wave is also one of the important factors in cavitation vortex dynamics. In terms of cavitation prediction methods, a detailed comparison and analysis were made between the traditional cavitation prediction methods based on numerical simulation and the currently popular cavitation prediction methods based on neural networks. The former mainly includes cavitation models and turbulence models, while the latter mainly summarizes the application of chain physics-informed neural network, pressure-velocity network, long short-term memory, and other neural networks in cavitation prediction. It is pointed out that artificial intelligence predictive models have advantages in model order reduction and accurate prediction of cavitation flow field feature parameters. In terms of cavitation suppression methods, active and passive cavitation suppression methods were thoroughly summarized. Finally, based on the current research status of hydrofoil cavitation prediction methods and cavitation suppression methods, this article discusses and looks forward to the direction of development.

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I. INTRODUCTION

Cavitation is a phenomenon prevalent in nature and is a significant issue in many hydropower engineering operations. When the local pressure of the liquid drops below the saturated vapor pressure, thus producing a large number of cavities. These bubbles continually expand and eventually annihilate,¹ and this series of processes is referred to as cavitation phenomenon. This phenomenon is widely observed in various machinery where liquid acts as a medium, such as hydrofoils,^{2,3} propellers,^{4–6} pumps,^{7–9} turbines,¹⁰ and other hydraulic machinery. Cavitation, as a complex two-phase flow phenomenon, primarily involves phase transition¹¹ and can be categorized based on its flow characteristics and forms observed in experiments into bubble cavitation,^{12–14} cloud cavitation,¹⁵ supercavitation,^{16–18} and vortex cavitation.^{19,20} The bubble cavitation stage is mostly a wandering bubble, with the expansion of the gas nucleus in its fluid

subsequently moving to the high-pressure region for annihilation. In the sheet cavitation stage, the cavitation appears in sheet form and attaches to the wall surface. After detachment, it progresses to cloud cavitation,²¹ where the detached bubbles appear as cloud structures, accompanied by increased vibration and noise in hydraulic machinery. In the supercavitation stage, bubbles significantly reduce the frictional resistance between the fluid equipment and the fluid. Vortex cavitation can be categorized into shear cavitation and vortex cavitation based on the vortex generation mechanism and its existence form. In the field of cavitation hydrodynamics, vortex cavitation generally occurs at the vortex core near the propeller blade tips.

Cavitation is a complex and unsteady flow phenomenon involving multiphase flow, heat and mass transfer, turbulence, and other physical processes. The cavitation process is usually accompanied by a series of unstable and adverse effects, such as increased vibration,^{22,23}

reduced performance,²⁴ enhanced noise, intensified pressure pulsations, and in severe cases, erosion of flow components, which can negatively impact the load-bearing capacity of equipment. Cavitation erosion can cause tiny damages to the surface of metal materials, which, when accumulated over time, may lead to fatigue and even structural failure. Furthermore, cavitation significantly impacts the energy efficiency and lifespan of hydraulic machinery such as ship propellers and water pumps.²⁵ For example, in ship propellers, cavitation erosion can cause the metal surface of the propeller to crack or peel off, affecting its performance and efficiency and shortening its service life.

In recent years, both domestic and international scholars have made some progress in understanding the cavitation evolution process and cavitation mechanisms.^{26–30} However, the instability mechanisms of cavitation remain a complex issue, with many aspects still warranting further investigation. Only through in-depth research, we can establish a robust theoretical foundation, enabling us to autonomously select appropriate suppression methods to control cavitation effectively. Moreover, cavitation prediction methods play a crucial role in cavitation research. Traditional numerical simulation methods for cavitation flows generally utilize homogeneous equilibrium models based on multiphase flow. The accuracy of cavitation predictions is significantly influenced by the choice of cavitation model and turbulence model. With the rapid development of artificial intelligence (AI) technologies such as machine learning, researchers have begun to apply these technologies to the results of computational fluid dynamics, which can, to some extent, address issues related to low numerical computation efficiency and high time costs. Therefore, this paper first delves into the mechanisms of cavitation, focusing primarily on the mechanisms of cavitation instability. Second, it thoroughly summarizes cavitation prediction methods, including traditional numerical simulation-based methods and neural network-based methods. Furthermore, to ensure the operational stability and performance of various hydraulic equipment, this paper primarily examines control methods aimed at mitigating cavitation hazards in hydrofoils and turbines, summarizing both active and passive cavitation suppression methods. Finally, the paper discusses the prospects and future directions for the mechanisms of cavitation flow suppression instability, cavitation prediction methods, and control methods.

II. CAVITATION FLOW INSTABILITY MECHANISMS

Among the various types of cavitation, cloud cavitation is one of the most hazardous, and it can induce multiple detrimental effects during its evolution. The formation of cloud cavitation is generally believed to result from the instability at the tail of sheet cavitation in the previous stage. Therefore, studying the mechanisms of cavitation instability can provide a theoretical foundation for subsequent research on cavitation prediction methods and suppression methods to mitigate cavitation hazards.

The damage caused by cavitation to hydraulic equipment is due to the implosion pressure generated by bubble collapse or the impact of high-speed fluid on the surface of flow components during the collapse phase.³¹ As illustrated in Fig. 1, it shows the cyclical process of reentrant jet and bubbly shock wave in a partially cavitation region.³² The adverse pressure gradient present at the tail of the cavity causes the fluid to move in the reverse flow direction, a phenomenon known as the reentrant jet. During the actual development of cavitation, there is a transformation between different types of cavitation. Multiple physical mechanisms, such as the reentrant jet mechanism and the

bubbly shock wave mechanism,³² can induce the phenomenon of sheet cavitation shedding.^{33,34} Cloud cavitation, in particular, induces multiple hazards during the evolution of cavitation, and the instability of cloud cavitation is a major driving force behind the negative impacts induced by cavitation. When cavitation progresses to the cloud cavitation stage, attached bubbles break off from the foil surface. Given a sufficiently large adverse pressure gradient, large-scale cloud cavitation clusters form and undergo periodic shedding and collapse.³⁵ The mechanisms of cloud cavitation shedding can be divided into reentrant jet and bubbly shock wave.

A. The reentrant jet impacts the cavitation flow mechanism

In early studies on cloud cavitation, researchers used high-speed photography to observe cloud cavitation, installed obstacles on hydrofoil surfaces to control cloud cavitation, and employed pressure transducers and hydrophones to measure the pressure. These methods contribute to understanding the mechanisms by which cloud cavitation is produced on hydrofoil sections. It was discovered that the shedding of sheet cavitation was caused by the reentrant jet originating from the rear edge of the sheet cavity,³⁶ thus recognizing that reentrant jet is the primary cause of cloud cavitation. The development of reentrant jet lead to instability in portions of the cavity. Measurements of pressure gradients obtained through laser Doppler vibrometry (LDV) have confirmed a strong correlation between high adverse pressure gradients and regions of cloud cavitation instability. The adverse pressure gradient also significantly impacts the thickness of reentrant jet. The reentrant jet is currently considered a major cause of cloud cavitation shedding. However, the mechanisms governing this physical phenomenon remain unclear. Additionally, given the complex structure of the reentrant jet, most previous research have explained mechanisms from a qualitative perspective, making it challenging to measure the flow field around the reentrant jet in detail. Gawandalkar and Poelma³⁷ used high-speed shadow photography to capture the dynamics of large-scale cavity shedding and employed tomographic imaging to measure the velocity and thickness of the reentrant jet. They discovered that the thickness of the reentrant jet relative to the cavity thickness is a critical factor affecting the shedding behavior of the cavity. The adverse pressure gradient and the cavity thickness compared to the thickness of the reentrant jet are also key parameters in analyzing the instability of the reentrant jet.³⁸

Some studies indicate that the collision between the reentrant jet and side jets plays a crucial role in both primary and secondary shedding.^{39–41} For instance, Ji *et al.*⁴² used a mass transfer cavitation model to numerically simulate and analyze the cavitation turbulence around a twisted hydrofoil. The study found a relationship between primary shedding and the collision of the radially diverging reentrant jet with the attached cavity surface. Additionally, it was discovered that the collision between the radially diverging reentrant jet and side jets directly leads to secondary cavity shedding. In a similar study, Movahedian *et al.*⁴³ conducted a large eddy simulation (LES) on sheet-cloud cavitation around a twisted 3D NACA16012 airfoil and observed the side jets induced by the reentrant jet. The direction of the reentrant jet and its distribution along the span were related to the structure of the airfoil itself. Furthermore, there was a correlation between the side jets and the radially diverging reentrant jet. The reentrant jet also caused primary cavity shedding, and the side jets led to secondary cavity

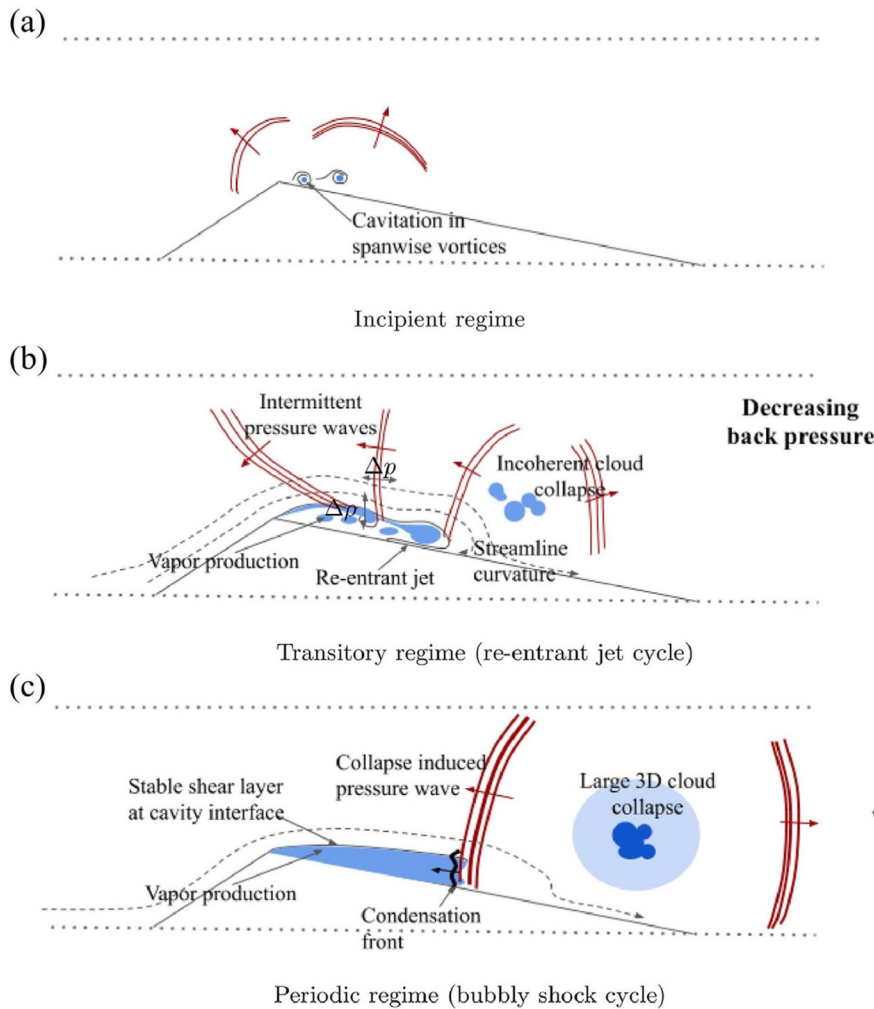


FIG. 1. The cyclical process of reentrant jet and bubbly shock wave in a partially cavitation region: (a) Incipient regime, (b) Reentrant jet cycle, and (c) bubbly shock cycle.³² Reproduced with permission from Int. J. Multiphase Flow **122**, 103155 (2020). Copyright 2019 Elsevier Ltd.

shedding. The adverse pressure gradient in the stagnation zone downstream of the cavity of the hydrofoil with a U-shaped flow structure forces the reentrant jet into the vapor structure,⁴⁴ thereby causing cavity shedding.

The effect of the reentrant jet on attached cavitation first appears in the cavitation development stage. During this phase, the reentrant jet intersects with the tail of the attached cavitation for the first time, resulting in a concave shape at the cavitation tail. In the cavitation maturation stage, the reentrant jet at the tail of the airfoil moves in the opposite direction to the incoming flow, moving close to the leading edge of the hydrofoil. The concavity at the tail of the cavitation deepens, and in the cavitation shedding phase, it comes into contact with the interface of the leading-edge bubble. The resulting shear force causes the cavitation to fracture and subsequently leads to the shedding of cloud cavitation. Meanwhile, newly formed bubbles reattach to the leading edge of the hydrofoil, and the hydrofoil surface cavitation enters the next cycle.⁴⁵ Figure 2 shows a comparison between experimental images and numerical simulation results of cavitation evolution,⁴⁶ from which it can be observed that large-scale bubbles shed under the influence of the reentrant jet. Thus, there is a close

connection between cavitation shedding and the reentrant jet, especially since the reentrant jet is one of the main causes of the shedding of cloud cavitation.

B. The bubbly shock wave impacts the cavitation flow mechanism

Another perspective on the shedding of cloud cavitation was proposed by Leroux, Ganesh, de Graaf, and Wu, among others. They argue that the unsteady shedding of cavitation, besides being influenced by the reentrant jet, also involves the formation and propagation of shockwaves, known as bubbly shock wave or condensation shock wave.^{47–52} The shedding of cloud bubbles from the previous cycle generates a bubbly shock wave, thereby initiating the shedding of the growing cavity,⁵³ with the bubbly shock wave acting over a shorter duration than the reentrant jet. Additionally, unlike the reentrant jet, the propagation of bubbly shock wave is not obstructed by surface obstacles. Most studies on cavitation flows are based on incompressible flow assumptions. However, Wang *et al.*⁵⁴ focused on compressible turbulent cavitation flows as their research subject, analyzing the

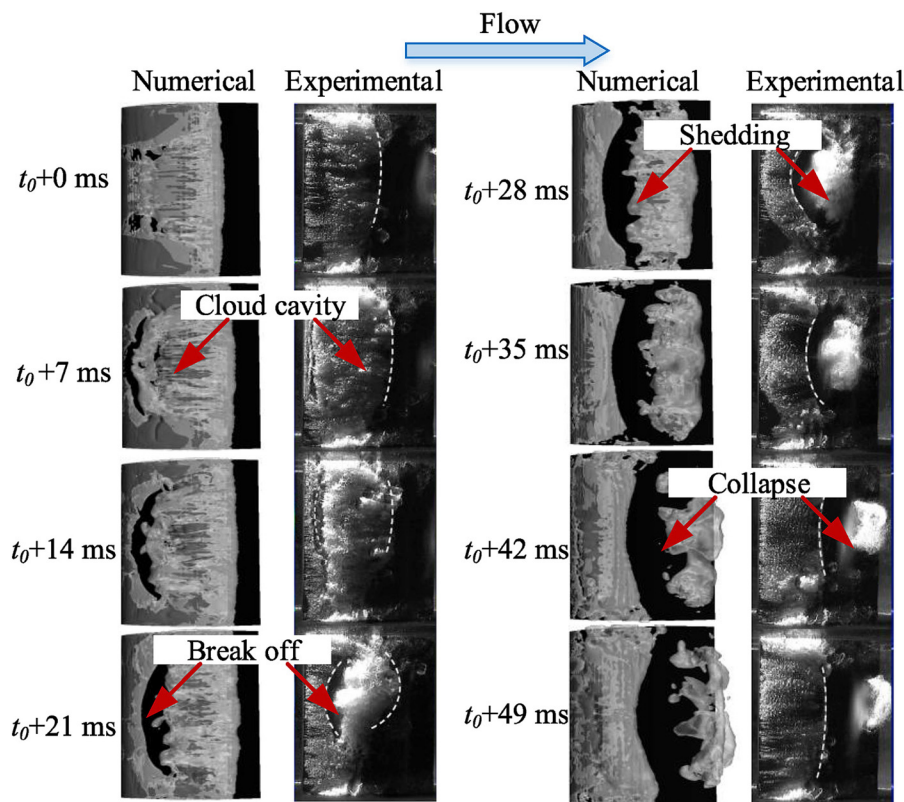


FIG. 2. A comparison between experimental images and numerical simulation results of cavitation evolution. Reproduced with permission from Sun *et al.*, *Ocean Eng.* **197**, 106831 (2020). Copyright 2019 Elsevier Ltd.⁴⁶

characteristics of typical cloud cavitation evolution. They proposed that the primary source of cavitation instability is the shockwave dynamics induced by cavity collapse. During the propagation phase of the shockwave, besides the reentrant jet, shockwave dynamics also play a significant role in cavitation vortex dynamics.⁵⁵ Cavitation typically refers to the formation of vapor regions around a liquid as it flows through hydraulic components, due to local reduction in pressure. The morphology of cavitation can change with variations in fluid velocity, local pressure, and temperature, making it a complex and dynamic phenomenon. During the instantaneous collapse of cloud cavitation, shockwaves are generated that can exert a certain impact force on the walls.⁵⁶ As the shockwaves propagate upstream, their high pressure causes the attached cavities in their path to contract sharply in the opposite direction of their initial development, with corresponding reductions in their length and bubble size. In reality, cavitation is a gas-liquid mixture; when the local bubble volume decreases, it leads to a reduction in the size of cavities within, further increasing the void fraction between local bubbles. The contraction of bubbles causes the cavitation to lose continuity, and upon reaching a certain level, this leads to bubble shedding. Afterwards, bubbles will begin to grow again, forming a quasi-periodic process.

The advancement of numerical simulation technology has enabled a better display and prediction of the mechanisms of the cavitation instability. Figure 3 provides a predictive view of the hydrofoil cavity shedding behavior under the reentrant jet and shockwave mechanisms.⁵⁷ From Fig. 3(a), it can be known that the reentrant jet developing upstream disrupts the stability of attached cavitation, forming

horseshoe vortex structures at the trailing edge of the foil. From Fig. 3(b), it can be seen that under the shockwave mechanism, there is no significant shedding behavior of surface cavities on the hydrofoil, but it leads to the attached cavities closely adhering to the foil surface and eventually collapsing. The reentrant jet caused by the adverse pressure gradient and the shockwaves induced by cavity collapse are factors influencing cavitation instability. However, in most current studies, the reentrant jet is considered to be the dominant factor in cavitation instability. At the same time, the role of shockwaves in cavitation flow has received increasing attention in recent years.

III. CAVITATION PREDICTION METHODS

The formation, shedding, and collapse of cavities constitute a complex dynamic process.⁵⁸ Throughout their development until collapse, they can impact hydraulic equipment and other flow-related components to varying degrees. To better investigate the mechanisms of cavitation flow instability, cavitation prediction methods have demonstrated their advantages in accurately predicting cavitation flow phenomena. Moreover, studying cavitation prediction methods provides valuable guidance for developing methods to mitigate the adverse effects of cavitation. Cavitation prediction methods can be summarized from two main perspectives: traditional cavitation prediction methods, which are based on numerical simulation methods, focusing on cavitation models and turbulence models; and the currently popular research methods, which are based on neural networks for cavitation prediction, including discussions on convolutional neural networks (CNNs), long short-term memory networks (LSTM), and others.

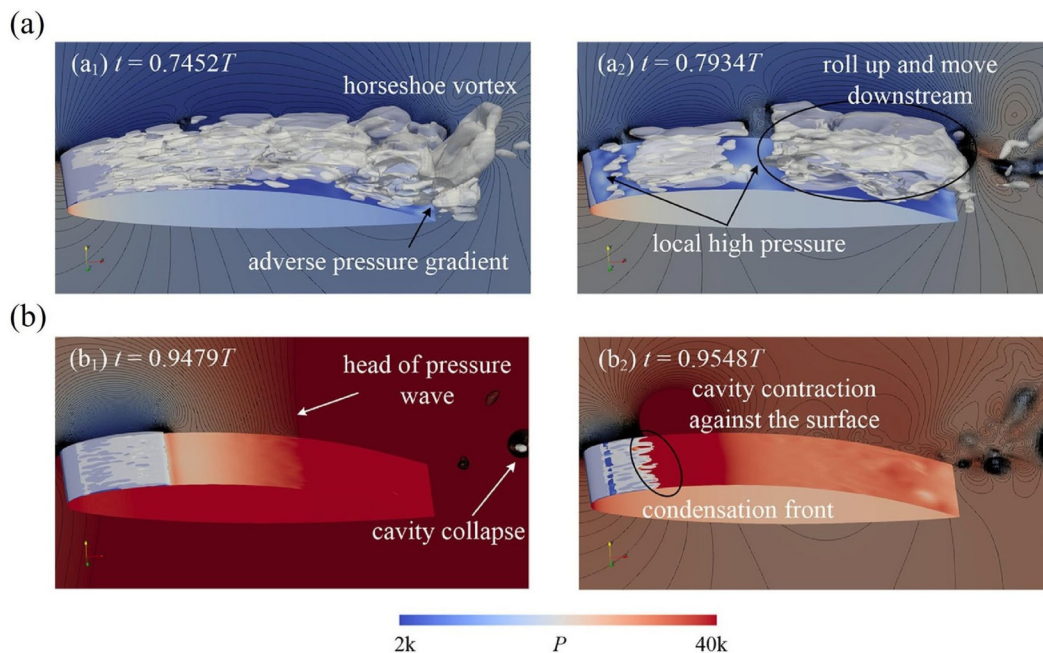


FIG. 3. A predictive view of the hydrofoil cavity shedding behavior under the reentrant jet and shockwave mechanisms: (a) Reentrant jet mechanisms, and (b) shockwave mechanisms. Reproduced with permission from Wang *et al.*, Appl. Math. Model. **100**, 410–431 (2021). Copyright 2021 Elsevier Ltd.⁵⁷

A. Cavitation prediction based on numerical simulation methods

In the early research of cavitation calculation models, potential flow theory was primarily utilized, and these models are still widely applied in various engineering applications.^{59,60} In recent years, with the improvement of computer hardware performance and the development of computational fluid dynamics (CFD), numerical simulation has become one of the important means to study cavitation issues, with CFD playing an increasingly significant role in cavitation research. The numerical simulation methods for cavitation, including cavitation models and turbulence models, play a key role in predicting cavitation phenomena. Cavitation, as a multi-scale complex unsteady flow phenomenon, relies significantly on the appropriate selection of cavitation models and turbulence models for numerical simulation, which in turn determines the predictive accuracy of cavitation phenomena to a certain extent.

1. Cavitation models

The cavitation flow phenomena in the flow field reveal significant unsteady characteristics, while also exhibiting the combined effects of gas and liquid convection characteristics. This is due to the substantial physical property differences between them, causing the gas–liquid surface to frequently exhibit highly unstable characteristics. Currently, the simulation accuracy of cavitation phenomena mainly relies on the establishment of models for gas–liquid phase change and mass transfer processes.⁶¹ In the numerical calculation methods for cavitation flow, the most commonly used approach is based on the homogeneous equilibrium model (HEM) for multiphase flow. HEM treats the two-phase mixture as a single fluid with variable density, utilizing the

weighted average of two-phase parameters as its flow parameters. It treats such multiphase flow as having the average characteristics of a single-phase fluid. The Kubota model,⁶² Merkle model,⁶³ Schnerr–Sauer cavitation model,⁶⁴ Kunz model,⁶⁵ Singhal model,⁶⁶ Zwart model,⁶⁷ etc. are all transport equation models based on HEM. Among these, the Zwart, Singhal, and Schnerr–Sauer models are based on the Rayleigh–Plesset (R–P) equation. Table I summarizes the source terms for these common cavitation models, along with some other models that have emerged in recent years.

Saberinia and Pasandidehfard⁶⁸ applied the Singhal, Zwart, Kunz, and Sauer cavitation models for the numerical simulation of cavitation flow around a Clark-Y airfoil. The results indicated that simulations with the Zwart cavitation model and a transient k - ϵ - ω transition turbulence model combination produced results that better matched experimental outcomes under the same conditions. Since the simulation accuracy of cavitation phenomena can be affected by the choice of empirical coefficients, cavitation models are continuously being improved. In addition, the cavitation patterns and cavitation dynamics are significantly influenced by the choice of evaporation and condensation coefficients in the Zwart cavitation model.⁶⁹ Jin *et al.*⁷⁰ addressed the adaptability issues of the evaporation and condensation coefficients in the Zwart model, developing a mathematical model that relates optimal coefficients with operational conditions and applied it to the quasi-steady cavitation flow around a NACA66 (MOD) hydrofoil. Compared to the original data, the accuracy of the coefficient predictions was effectively improved by 12%. Furthermore, after considering the effects of vortex factors in the mass transfer process, the improved cavitation model predicted cavitation patterns that matched experimental results more closely than the original Zwart model. Model predictions showed that a decrease in cavitation number leads to the

TABLE I. The source terms for some common cavitation models.

Main contributors and years	Evaporation source term $\dot{m}^+$ and condensation source term $\dot{m}^-$	References
Kubota, 1992	$\begin{cases} \dot{m}^+ = C_{\text{vap}} \frac{3\alpha_v \rho_v}{R_b} \sqrt{\frac{2(p-p_v)}{3\rho_l}} \\ \dot{m}^- = C_{\text{cond}} \frac{3\alpha_{\text{nuc}}(1-\alpha_v)\rho_v}{R_b} \sqrt{\frac{2(p-p_v)}{3\rho_l}} \end{cases}$	62
Merkle, 1998	$\begin{cases} \dot{m}^+ = \frac{C_{\text{vap}} \rho_v \max(0, p-p_v)(1-\alpha_l)}{0.5\rho_l U_\infty^2 t_\infty} \\ \dot{m}^- = \frac{C_{\text{cond}} \rho_l \min(0, p-p_v)\alpha_l}{(0.5\rho_l U_\infty^2) t_\infty} \end{cases}$	63
Sauer, 2000	$\begin{cases} \dot{m}^+ = \frac{3\rho_v \rho_l \alpha_v^2}{R_b \rho_m} \sqrt{\frac{2(p-p_v)}{3\rho_l}} \\ \dot{m}^- = \frac{3\rho_v \rho_l \alpha_v(1-\alpha_v)}{R_b \rho_m} \sqrt{\frac{2(p-p_v)}{3\rho_l}} \end{cases}$	64
Kunz, 2000	$\begin{cases} \dot{m}^+ = \frac{C_{\text{vap}} \rho_v (\alpha_l - \alpha_{\text{ng}})^2 (1 - \alpha_l - \alpha_{\text{ng}})}{t_\infty} \\ \dot{m}^- = \frac{C_{\text{cond}} \rho_v \alpha_l \min(0, p-p_v)}{(0.5\rho_l U_\infty^2) t_\infty} \end{cases}$	65
Singhal, 2002	$\begin{cases} \dot{m}^+ = \frac{C_{\text{vap}} \sqrt{k} \rho_l \rho_v}{\sigma} \sqrt{\frac{2(p-p_v)}{3\rho_l}} (1 - f_v - f_g), & p \leq p_v \\ \dot{m}^- = \frac{C_{\text{cond}} \sqrt{k} \rho_l \rho_l}{\sigma} \sqrt{\frac{2(p-p_v)}{3\rho_l}} f_v, & p > p_v \end{cases}$	66
Zwart, 2004	$\begin{cases} \dot{m}^+ = C_{\text{vap}} \frac{3\alpha_{\text{nuc}}(1-\alpha_v)\rho_v}{R_b} \sqrt{\frac{2(p_v-p)}{3\rho_l}}, & p \leq p_v \\ \dot{m}^- = C_{\text{cond}} \frac{3\alpha_v \rho_v}{R_b} \sqrt{\frac{2(p-p_v)}{3\rho_l}}, & p > p_v \end{cases}$	67
Huang, 2010	$\begin{cases} \dot{m}^+ = \chi(b) C_{\text{vap}} \frac{3\alpha_v \rho_v}{R_b} \sqrt{\frac{2(p-p_v)}{3\rho_l}} + (1-\chi(b)) \frac{\rho_l^2 \min(p-p_v, 0) \alpha_l}{\rho_v (V_{v,n} - V_l)^2 (\rho_l - \rho_v) t_\infty}, & p \leq p_v \\ \dot{m}^- = \chi(b) C_{\text{cond}} \frac{3\alpha_v}{R_b} \sqrt{\frac{2(p-p_v)}{3\rho_l}} + (1-\chi(b)) \frac{\rho_l \max(p-p_v, 0) \alpha_v}{(V_{v,n} - V_l)^2 (\rho_l - \rho_v) t_\infty}, & p > p_v \end{cases}$ $\chi(b) = 0.5 + \tanh \left[\frac{C_1(0.6b - C_2)}{0.2(1 - 2C_2) + C_2} \right] / 2 \tanh(C_1)$ $b = \rho_m / \rho_l$	80
Kadivar, 2018	$\begin{cases} \dot{m}^+ = \frac{\rho_v \rho_l}{\rho_m} \alpha_v (1 - \alpha_v) \frac{3}{R_b} \sqrt{\frac{2 \max(p-p_v, 0)}{3\rho_l}} \\ \dot{m}^- = \frac{\rho_v \rho_l}{\rho_m} \alpha_v (1 - \alpha_v) \frac{3}{R_b} \sqrt{\frac{2 \max(p-p_v, 0)}{3\rho_l}} \end{cases}$	81

TABLE I. (Continued.)

Main contributors and years	Evaporation source term $\dot{m}^+$ and condensation source term $\dot{m}^-$	References
Shi, 2021	$\text{sign}(p_v - p) \frac{3\alpha_v \rho_l \rho_v}{R \rho_m} \sqrt{\frac{2(p_v - p)}{3\rho_m} \left[1 - \left(\frac{R_0}{R} \right)^3 \right] - \frac{2S}{\rho_m} \frac{(R^2 - R_0^2)}{R^3}}$ $R = \left(\frac{3\alpha_v}{4n\pi} \right)^{\frac{1}{3}}, R_0 = \left(\frac{3\alpha_0}{4n\pi} \right)^{\frac{1}{3}}$ $\dot{m}^+ = C_{\text{vap}} (4\pi N)^{\frac{1}{3}} (3\alpha_v)^{\frac{2}{3}} (1 - \alpha_v)^{\frac{4}{3}} \frac{\rho_l \rho_v}{\rho_m} A$ $\times \left(\sqrt{\left(\frac{p_v(T) - p}{\text{NPSHa}} + \frac{T_\infty - T_v}{\Delta T_{\text{sub}}} \right) + \frac{A^2}{B^2} t} - \sqrt{\frac{A^2}{B^2} t} \right)$ $\dot{m}^- = C_{\text{cond}} (4\pi N)^{\frac{1}{3}} (3\alpha_v)^{\frac{2}{3}} (1 - \alpha_v)^{\frac{4}{3}} \frac{\rho_l \rho_v}{\rho_m} A$ $\times \left(\sqrt{\left(\frac{p - p_v(T)}{\text{NPSHa}} + \frac{T_\infty - T_v}{\Delta T_{\text{sub}}} \right) + \frac{A^2}{B^2} t} - \sqrt{\frac{A^2}{B^2} t} \right)$ $A = \sqrt{\frac{2}{3} \frac{\text{NPSHa}}{\rho_l}}, \text{NPSHa} = p_\infty - p_v(T_\infty)$ $B = \sqrt{\frac{12}{\pi} \mathcal{D} J}, J = \frac{\rho_l c_{\text{pl}} \Delta T_{\text{sub}}}{\rho_v \mathcal{L}}, \Delta T_{\text{sub}} = T_{\text{sat}} - T_\infty$	77
Li, 2023		78

gradual extension of the leading-edge cavity to the suction side, which in turn results in a reduction of lift and tip-leakage vortex (TLV) for the hydrofoil.⁷¹

Combining large eddy simulation (LES) with the Schnerr–Sauer cavitation model can effectively capture the characteristic of cavitation structures of sheet/cloud cavitation and the transient dynamic behavior of cavitation flow.^{72–74} Figure 4 shows experimental images and corresponding numerical simulations of a NACA66 (MOD) airfoil at different moments during the transition stage of sheet/cloud cavitation and cloud cavitation.⁷² The error in the cavitation cycle is around 5%, and from the figure, it can be seen that the cavitation evolution patterns were well predicted. In addition, by considering non-condensable gases and using an improved Schnerr–Sauer cavitation model⁷⁵ combined with LES to simulate the cavitation flow around a hydrofoil, effective prediction of pressure fluctuations on the hydrofoil surface can be achieved. Hong *et al.*⁷⁶ proposed a new truncated form of the cavitation model based on the Rayleigh–Plesset (R–P) equation, which could control the source terms for vapor generation and destruction. This model was compared with the Schnerr–Sauer cavitation model. The results indicated that the proposed model has sufficient robustness and can accurately predict the evolution process of sheet/cloud cavitation, aligning with the quantitative data characteristics observed in experiments.

Considering the interactions between bubbles, Shi *et al.*⁷⁷ coupled the bubble growth rate with the mass transfer equation between the vapor and liquid phases in their work. Moreover, to calculate the bubble fraction in multi-cavitation regions under different cavitation conditions, Li *et al.*⁷⁸ utilized linear interpolation along the saturation

pressure–temperature curve to replace the Clausius–Clapeyron equation. These models enhance the simulation accuracy of cavitation phenomena and, to some extent, provide guidance for subsequent related research.

It can be seen that by correcting cavitation models, such as incorporating non-condensable terms, adding vortex factors, and adaptively adjusting model empirical coefficients, the prediction accuracy of cavitation models regarding characteristics of the cavitation flow field can be enhanced to some extent. This, in turn, provides more accurate data support for exploring cavitation evolution mechanisms and cavitation control strategies. Additionally, turbulence kinetic energy and other factors can also be included in the correction methods.⁷⁹

2. Turbulence models

Turbulence models are employed to predict the oscillatory behavior of cavitation flow. The numerical simulation methods for turbulence models commonly in use today can be primarily categorized into three main types: direct numerical simulation (DNS), scale-resolving simulation (SRS), and Reynolds average Navier–Stokes (RANS). Each of these numerical simulation methods for turbulence models has its own set of advantages and disadvantages, as illustrated in Fig. 5.

DNS can directly solve the unsteady Navier–Stokes equations for fluid flow and is typically considered the most accurate simulation method, offering high precision and capturing the most comprehensive flow field information. However, the computational cost of DNS is extremely high, increasing with the cube of the Reynolds number, which limits its practical engineering applications. The SRS method

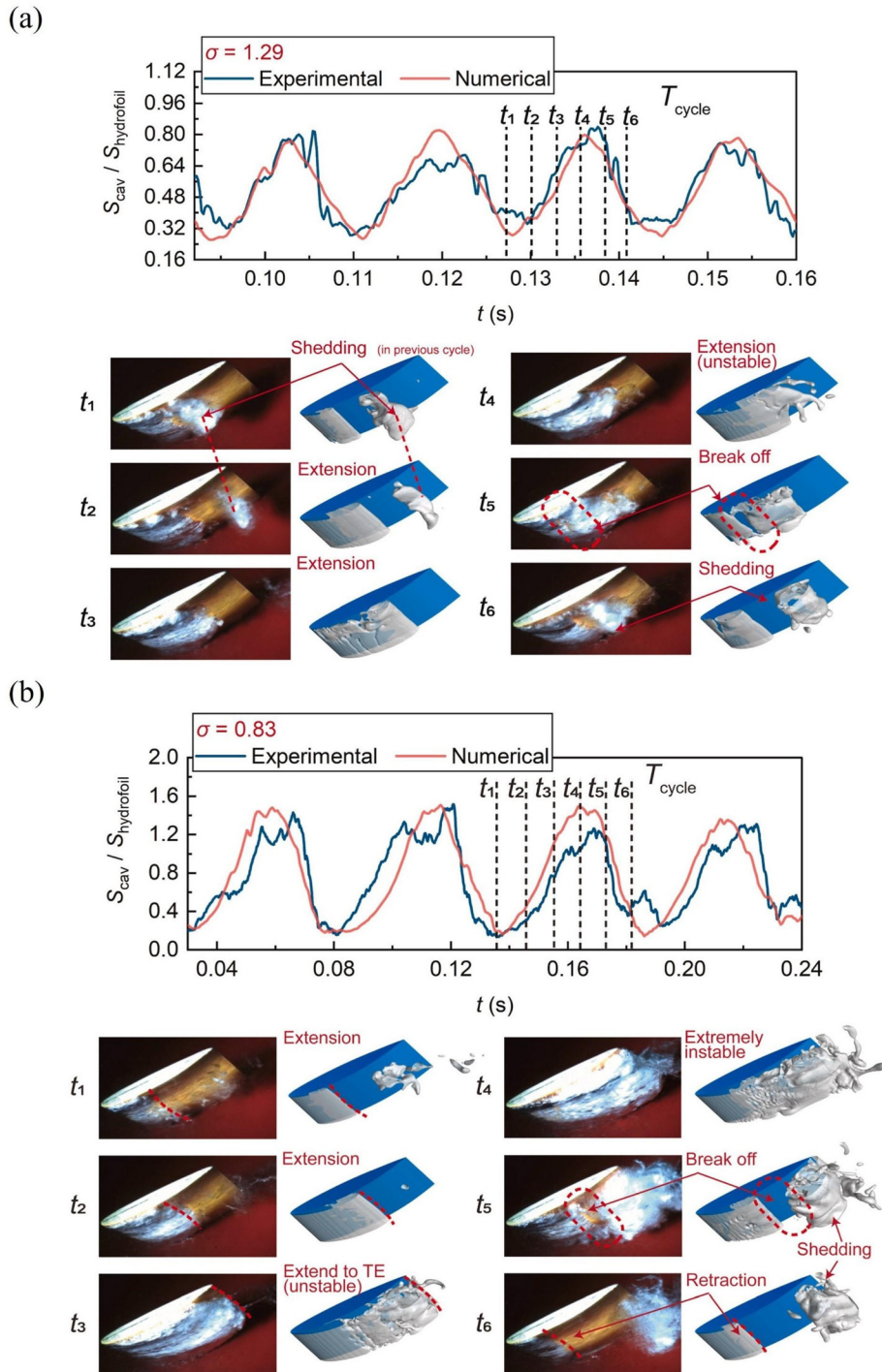


FIG. 4. Periodic curves and numerical results of dimensionless cavitation regions ($S_{cav}(t)/S_{hydrofoil}$) over a period of time under different cavitation numbers: (a) Periodic curves of experimental data and numerical simulation results, and (b) experimental images and corresponding numerical simulation images of the water wing at different times from t_1 to t_6 . $S_{hydrofoil}$: the cross section area of the hydrofoil, S_{cav} : cavitation area. Reproduced with permission from Li et al., Ocean Eng. **280**, 114584 (2023). Copyright 2023 Elsevier Ltd.⁷²

directly resolves some of the turbulent structures in the flow field, while the remaining turbulence is computed using mathematical models.⁸² Common models include large eddy simulation (LES), scale-adaptive simulation (SAS), and detached eddy simulation (DES). Moreover, SRS has an advantage over RANS in capturing the transient

characteristics of the flow.⁸³ Due to the need for relatively small time steps for simulation calculations, the computational resources consumed by SRS are much greater compared to RANS.

Although DNS and SRS methods offer more accuracy in computational precision compared to RANS, they require significantly more

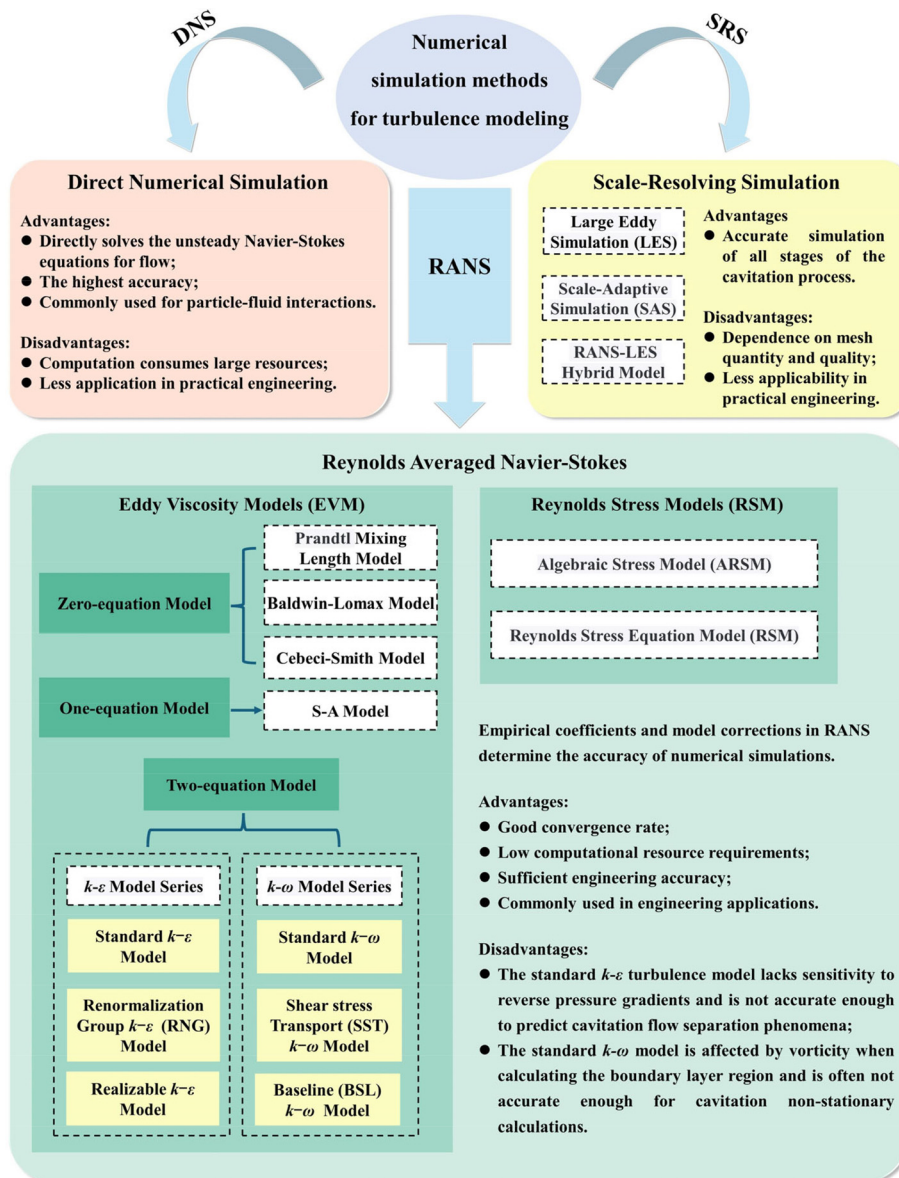


FIG. 5. Numerical simulation methods and advantages and disadvantages of turbulence models.

computational resources and are less frequently applied in practical engineering projects. RANS, due to its higher engineering accuracy, faster convergence, and lower consumption of computational resources, continues to be popular in modern industrial applications. Common two-equation Reynolds-averaged models in RANS include the k - ε model and the k - ω model. In 1986, Yakhot and Orszag⁸⁴ proposed the RNG k - ε model, with the equation for calculating the turbulent viscosity μ_t given as follows:

$$\mu_t = C_\mu \rho_m \frac{k^2}{\varepsilon}, \quad (1)$$

where C_μ is a constant, $C_\mu = 0.09$; k is the turbulent flow energy generation term; ε is the turbulent dissipation term; and ρ_m is the density of the gas-liquid mixture phase.

Menter⁸⁵ proposed the initial SST k - ω model (Menter's SST k - ω) in 1994, where the calculation formula for turbulent viscosity μ_t is given as follows:

$$\mu_t = \frac{a_1 \rho_m k}{\max(a_1 \omega, \Omega F_2)}, \quad (2)$$

$$F_2 = \tanh(\arg_2^2), \quad (3)$$

$$\arg_2 = \max\left(2 \frac{\sqrt{k}}{0.09 \omega y}, \frac{500 \nu}{y^2 \omega}\right), \quad (4)$$

where a_1 is the model constant, $a_1 = 0.31$; ω is the specific dissipation rate; Ω is the absolute value of vorticity; F_2 is a mixed function in the

form of tangent function; y is the closest distance to the next surface, m; ν is the kinematic viscosity, Pa s.

A decade after the SST k - ω model was applied in industry, in 2003, Menter *et al.*⁸⁶ made a correction to the calculation of the turbulent viscosity μ_t within the SST k - ω model. They replaced the vorticity Ω with the strain rate's invariant measure S , while all other parameters remained the same. The corrected equation for calculating the turbulent viscosity μ_t is as follows:

$$\mu_t = \frac{a_1 \rho_m k}{\max(a_1 \omega, SF_2)}. \quad (5)$$

Compared to the RNG k - ε model, the SST k - ω model does not require the use of wall functions to enhance simulation accuracy in the near-wall region. By introducing the transport equations for both turbulent kinetic energy k and specific dissipation rate ω , the SST k - ω model utilizes the k - ω model in near-wall regions to demonstrate its excellent accuracy in simulating turbulent flows. Furthermore, by introducing a blending factor, it combines the original k - ε and k - ω models. This approach allows for the use of the k - ε model further away from the wall (at the boundary layer edge and free shear layers),

combining the advantages of both models. It also addresses the issue of the k - ω model being overly sensitive to free-stream turbulence. As a result, the SST k - ω model is capable of effectively handling both near-wall turbulent flow and free-stream turbulent flow away from the wall, significantly enhancing the stability of the algorithm.

Additionally, standard turbulence models usually have the drawback of over-predicting the turbulence viscosity coefficient when numerically simulating the cavitation flow process. Therefore, it is often necessary to correct the turbulence models to enhance the accuracy of the numerical simulation results. Table II summarizes several corrections to turbulence models. These corrected models can reduce the turbulence viscosity to some extent, thereby improving the ability to represent the characteristics of cavitation flow, and further enhancing the simulation accuracy of predicting cavitation. Currently, there are several main methods to correct the turbulence models:

a. Filter-based model (FBM). Johansen *et al.*⁸⁷ introduced a parameter based on filter size into the standard k - ε model, greatly improving its original prediction capability. The filter turbulence viscosity $\mu_{t\text{-FBM}}$ is shown as follows:

TABLE II. Some corrected turbulence models.

Main contributors and years	Turbulent viscosity correction equation	Model	Effect	References
Coutier Delgosha, 2003	$\mu_t = f(\rho) \frac{C_\mu k^2}{\varepsilon}, C_\mu = 0.085$ $f(\rho) = \rho_v + \left(\frac{\rho_m - \rho_l}{\rho_v - \rho_l} \right)^n (\rho_l - \rho_v)$	RNG k - ε model	The actual viscosity of the mixture is reduced, taking into account the effect of the high compressibility of the gas-liquid mixture on the turbulent flow structure.	99
Zhang, 2017	$\mu_t = C_\mu \frac{\rho k^2}{\varepsilon} F$ $F = \min \left[f(n), C_3 \frac{\Delta}{l_{\text{RANS}}} \right]$	DCM and FBM hybrid model	It reduced turbulent viscosity and limited over-prediction of cavitation flow on hydrofoil walls and in wake flow.	100
Chen, 2019	$\mu_t = f(\rho) \frac{C_\omega k}{\omega}, C_\omega = \frac{1}{\max \left(\frac{1}{a^*}, \frac{SF_2}{a_1 \omega} \right)}$ $f(\rho) = \rho_v + \frac{(\rho_m - \rho_v)^n}{(\rho_l - \rho_v)^{n-1}}$	SST k - ω model	It effectively solved the problem of failing to accurately predict cavitation shedding due to over-prediction of turbulent viscosity in the cavitation wake during cavitation flow.	101
Gu, 2022	$\mu_{\text{DCMFBM}} = C_u \frac{\rho_m k^2}{\varepsilon} [\phi f_{\text{FBM}} + (1 - \phi) f_{\text{DCM}}]$ $f_{\text{DCM}} = \rho_v + \frac{(\rho_m - \rho_v)^n}{(\rho_l - \rho_v)^{n-1}}$ $f_{\text{FBM}} = \min \left(1, \frac{\Delta \varepsilon}{k^2} \right)$ $\phi = 0.5 + \frac{\tanh \left[\frac{C_1 (0.6 \rho_m / \rho_l - C_2)}{0.2 (1 - 2C_2) + C_2} \right]}{2 \tanh C_1}$	DCM and FBM hybrid model	It more accurately matched the characteristic parameters of the flow field and the shape of the cavities in the cavitation process.	61
Song, 2023	$f(\rho) = \rho_v + \left(\frac{\rho_v - \rho_m}{\rho_v - \rho_l} \right)^{10} (\rho_l - \rho_v)$ $\mu_{t,m} = f(\rho) + \frac{\alpha_1 k}{\max(\alpha_1 \omega, SF_2)} \rho_v$	SAS-SST model	It greatly reduced the error precision of numerical simulation and test results of gas-liquid two-phase flow and controlled it within 0.3%.	102

$$\mu_{t-FBM} = C_\mu \rho_m f_{FBM} \frac{k^2}{\varepsilon}. \quad (6)$$

The filtering function f_{FBM} is as follows:

$$f_{FBM} = \min \left(1, C_1 \frac{\Delta \varepsilon}{k^{3/2}} \right), \quad (7)$$

where C_1 is a constant, $C_1=1.0$; Δ is the filter size.

In study,⁸⁷ it was found that in all coarse, medium, and fine meshes examined, FBM demonstrates better simulation results than the standard $k-\varepsilon$ model, and as the size of the filter increases, this model approximates the standard $k-\varepsilon$ model quite well. Zhang *et al.*⁸⁸ proposed an improved FBM and compared to the original FBM and standard $k-\varepsilon$ model, the improved model can more accurately predict the shedding frequency of unsteady cloud cavitation and the pressure coefficient at the midplane. The experimental results and transient cloud cavitation structure predicted by the improved FBM are in good agreement. In addition, compared to the original FBM, the turbulence model based on curvature correction filtering (FBM-CC) proposed using the Spalart–Shur correction term has made great improvements in calculation accuracy and efficiency, and has reduced the flow energy in the area near the vortex axis. This model can be applied to the numerical simulation of turbulent curved flow in the engineering field.⁸⁹

b. Density correction model (DCM). In order to improve the simulation accuracy of the cavitation flow, a local density correction is applied to the turbulent eddy viscosity, with the equation for calculating the turbulent viscosity μ_{t-DCM} ⁹⁰ given as follows:

$$\mu_{t-DCM} = \frac{f(\rho) C_\mu k^2}{\varepsilon}. \quad (8)$$

The density correction function $f(\rho)$ is as follows:

$$f(\rho) = \rho_v + (1 - \alpha_v)^n (\rho_l - \rho_v), \quad (9)$$

where α_v is the volume fraction of the vapor phase; ρ_l is the density of the liquid; ρ_v is the density of the vapor.

Furthermore, to improve the numerical simulation accuracy of the RNG $k-\varepsilon$ turbulence model, an improved turbulence closure density correction model (MDCM) was used in the study of cavitation flow on Clark-Y hydrofoil. Both DCM and MDCM were able to accurately capture the change trend of the drag coefficient of the hydrofoil in the cloud cavitation phase.⁹¹

c. Mixed turbulence model. In order to reduce the turbulent eddy viscosity during the cavitation flow process, Huang *et al.*⁹² mixed FBM and DCM and proposed the filter-based density correct model (FBDCM), which takes full advantage of the strengths of both models and can accurately model the cavitation areas near wall and far from wall. The corrected mixed turbulent viscosity μ_{FBDCM} is as follows:

$$\mu_{FBDCM} = \frac{C_\mu \rho_m k^2}{\varepsilon} f_{FBDCM}, \quad (10)$$

$$f_{FBDCM} = \varphi f_{FBM} + (1 - \varphi) f_{DCM}, \quad (11)$$

where C_2 is a constant, $C_2=4$; φ is the mixing function.

Additionally, in some studies,^{93–96} the FBDCM was also used to carry out numerical simulations of cavitation dynamics. Through this

hybrid model, the numerical simulation results matched the experimental data well, indicating that this model can better predict unsteady cavitation processes and capture flow field characteristics. In the study of unsteady cavitation flow on hydrofoils, the use of this model has been able to accurately predict transient pressure pulsations and bubble shedding structures,⁹⁷ and the initial cavitation, the development of cavitation, and the shedding of cavitation can be better captured.⁹⁸ Numerical simulation results are close to experimental results, providing a more accurate prediction of the cavitation evolution process on hydrofoils.

By correcting the turbulent viscosity terms in the turbulence models, it is possible to more accurately simulate the cavitation flow process, which in turn allows for a more accurate capture of the shapes and multi-scale vortex structures of the bubbles at various stages on the surfaces of hydrofoils and other hydraulic equipment.

B. Cavitation prediction based on neural networks

Cavitation prediction based on the numerical simulation method usually obtains the pressure distribution, velocity field distribution, and surface pressure coefficients of hydrofoils and other hydraulic equipment through numerical simulation or experiments. However, traditional methods have problems such as large computation and high time cost. In recent years, artificial intelligence has developed rapidly, and how to apply artificial intelligence into traditional methods has become an urgent problem for researchers to solve. In order to make full use of the numerical simulation results of computational fluid dynamics and improve the computational efficiency of numerical simulation, various artificial intelligences such as machine learning, deep learning, and neural networks have been integrated into traditional technologies. The principle is that by importing single or multiple parameters into the pre-established mathematical model, and after a series of simple/complex processing, the prediction model is finally obtained. This article summarizes the use of the currently popular neural network model as the basis for the cavitation prediction model.

Neural networks can be divided into feedforward neural network and feedback neural network based on whether there is feedback for the input information. Figure 6 is a conceptual diagram of a neural network based on whether the input information has feedback. In the feedforward neural network [Fig. 6(a)], information starts at the input layer, then passes through the neurons of each layer in sequential order until it is transmitted to the output layer. The entire network does not provide feedback processing for the information transfer process, common feedforward neural networks include convolutional neural networks (CNN), back propagation (BP) neural networks, generative adversarial networks (GAN), radial basis function (RBF) neural networks, etc. Long short-term memory (LSTM), recurrent neural networks (RNN), and Hopfield networks are classic feedback neural networks. As shown in Fig. 6(b), in these types of neural networks, neurons can also receive feedback signals from themselves, giving them memory function, and the same neuron can display different states at different time points. Information transmission in feedback neural networks can be either one-way or two-way.

Due to the high time cost and computational consumption of traditional computational fluid dynamics (CFD) for complex models, models based on neural networks are increasingly used by researchers for rapid prediction of cavitation flow fields. Zhang *et al.*¹⁰³ set up pressure monitoring points at the tip and mid positions of the

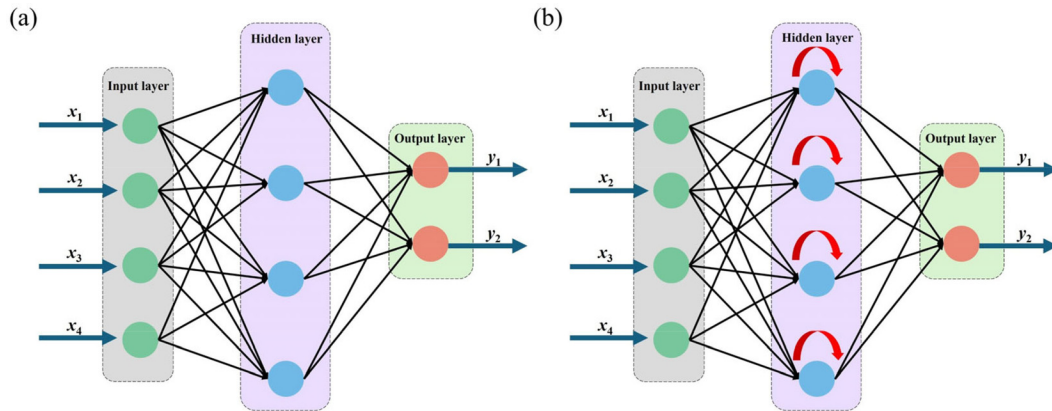


FIG. 6. A conceptual diagram of a neural network based on whether the input information has feedback: (a) feedforward neural network and (b) feedback neural network.

hydrofoil and constructed a hydrofoil surface pressure prediction model for multivariate time series based on the long short-term memory (LSTM) network, using horizontal speed, vertical speed, and gas phase volume fraction as inputs for the model. The results show that the pressure data on the hydrofoil surface predicted by the model can fit well with the data obtained by LES, and it has certain generalization ability. In order to enhance the ability to capture multi-scale features of the flow field, Xu *et al.*¹⁰⁴ proposed the pressure-velocity network (PVNet) based on the improved U-shaped neural network, replacing the last encoding module of UNet with Transformers [right side of Fig. 7(a): Architecture of PVNet], and predicted the velocity field through 12 pressure sampling points on the hydrofoil surface. The input of the dataset is the numerical results of different pressure sampling points under different cavitation numbers and attack angles in the cavitation flow of the NACA66 hydrofoil, and the distance from different positions in the flow field to the blunt body is encoded to represent its flow characteristics, the black arrow in the figure indicates the data transmission direction [left side of Fig. 7(a): Architecture of UNet]. From the prediction of horizontal and vertical velocity on the surface of the hydrofoil by different models [Fig. 7(a)], it can be seen that in the three neural networks of shallow neural network (SNN), UNet, and PVNet, their predicted velocity fields basically match the results of numerical simulations. However, SNN performs relatively weakly in areas with large velocity gradients compared to the other two models, while PVNet can capture velocity changes that UNet cannot capture, thus making it possible to implement the reconstruction of the flow field with high precision.

In addition, model reduction methods can better observe high-dimensional problems from a low-dimensional perspective, thereby achieving the identification of coherent structures in high-dimensional fluid dynamics systems. Li *et al.*¹⁰⁵ proposed an optimized hydrofoil framework based on deep learning, which is used to predict the pressure field distribution of hydrofoils under specific operating conditions, realizing the prediction between hydrofoil geometry, parameter design, and hydrodynamic performance. Figure 7(b) is a hydrofoil optimization framework based on deep learning. It adopts pressure field reconstruction and hydrodynamic performance prediction network based on CNN, uses them as surrogate models, and combines them with multi-objective optimization algorithms, and then performs

numerical simulations on the optimized hydrofoil models obtained. Case 1 and case 2, respectively, display the numerical simulation and prediction results of the pressure coefficient on the surface of the hydrofoil. There is no significant difference between the numerical simulation results and the predicted pressure coefficient results, and the average relative error of the prediction results is 6.83%, reflecting the effectiveness of the pressure field reconstruction model. Ouyang *et al.*¹⁰⁶ were the first to apply neural networks based on physical information to the study of unsteady cavitation flow, and on this basis proposed a chain physics-informed neural network (PINN) [Fig. 7(c)], using the NACA66 hydrofoil to predict its surface velocity field, pressure field, and other cavitation flow field parameters. The input of the model is a sparse set of different velocity components $[u, v, w]$ obtained from experiments. During the prediction process of the model, smaller scale cavities merge with larger scale cavities, causing the length predicted by the chain PINN in terms of details to be relatively slightly longer, but overall, the cavitation length predicted by the chain PINN approximately matches the experiments and DNS results. The abrupt change in gradient caused by cavitation fracture and shedding in the local area makes the chain PINN unable to predict small-scale cavitation well. However, chain PINN can effectively predict the growth rate of the cavity, the moment of fracture, etc., while the error between the predicted values and the experimental values in the pressure field, horizontal split speed, and liquid fraction field are all within 0.4%, achieving accurate prediction of relevant characteristic parameters during the cavitation process.

Xu *et al.*¹⁰⁷ studied the Clark-Y hydrofoil model and proposed a new machine learning assisted RANS method for unsteady bubble flow around the hydrofoil. The results indicate that the proposed new method can better characterize the shedding process of sheet cavitation compared to the original RANS results. In addition, optimizing machine learning can improve the accuracy of predicting cavitation characteristics. Kamali *et al.*¹⁰⁸ studied the effect of different cavitation angles on ventilation cavity characteristics under low ventilation coefficients. The random forest (RF) and extreme gradient boosting (XGB) models can be optimized using Bayesian optimization algorithm (BOA). The results indicate that compared with the RF model, the BOA-XGB model has appropriate accuracy in predicting cavitation characteristics.

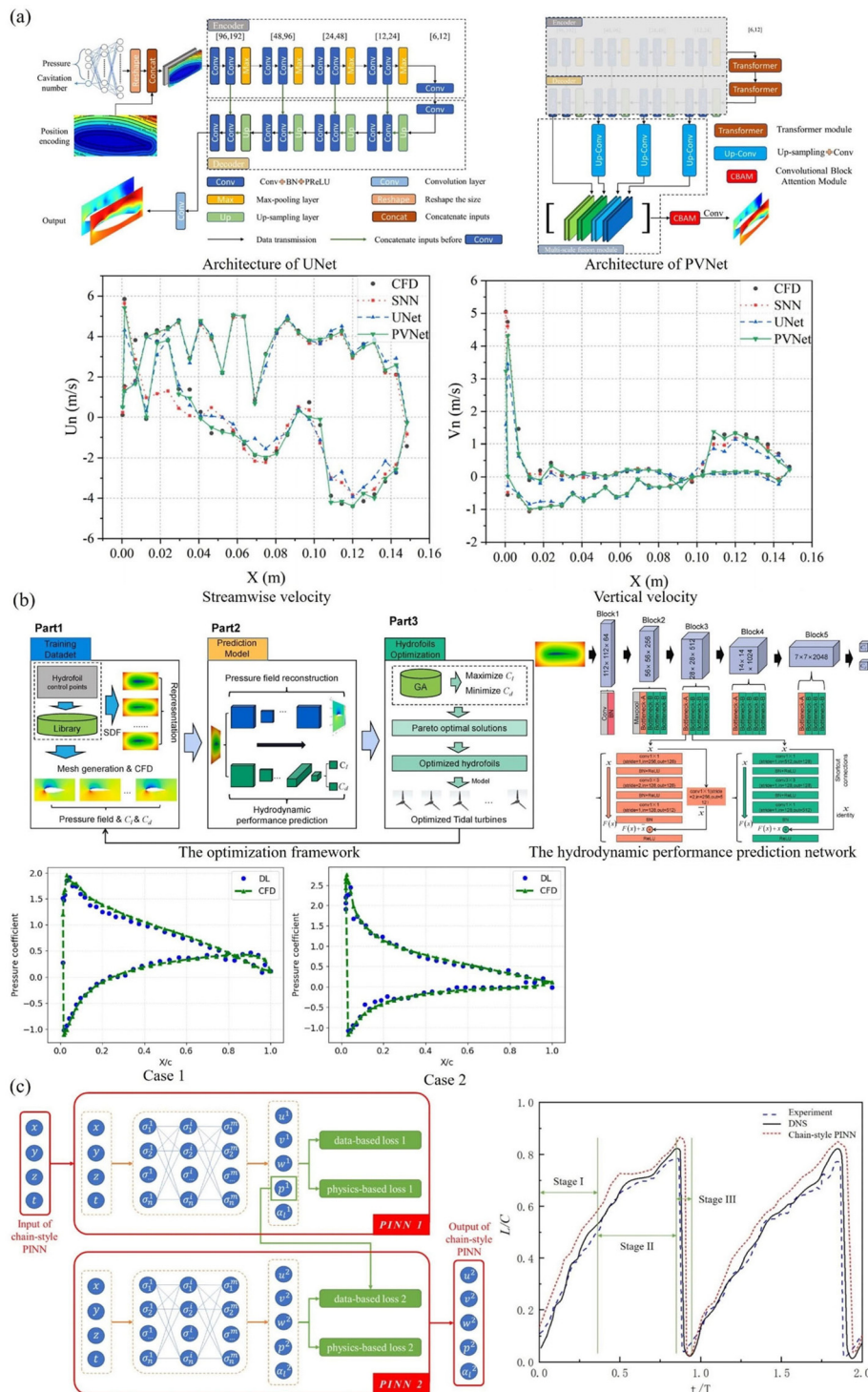


FIG. 7. Comparison of prediction models and results of different neural networks: (a) UNet and PVNet. Reproduced from Xu *et al.*, Phys. Fluids 35, 083323 (2023) with the permission of AIP Publishing LLC;¹⁰⁴ (b) Deep learning and CNN. Reproduced with permission from Li *et al.*, Ocean Eng. 305, 117996 (2024). Copyright 2024 Elsevier Ltd.;¹⁰⁵ and (c) chain PINN. Reproduced with permission from Eng. Appl. Artif. Intell. 119, 105724 (2023). Copyright 2022 Elsevier Ltd.¹⁰⁶

In summary, both numerical simulation-based and neural network-based cavitation prediction methods play indispensable roles in their respective fields. The selection of appropriate cavitation and turbulence models, or modified models, is crucial for accurately predicting

cavitation flow phenomena. Meanwhile, machine learning techniques, such as neural networks, have gained increasing attention in recent years. By integrating with traditional numerical simulation methods, they can significantly enhance the efficiency of cavitation predictions.

IV. CAVITATION SUPPRESSION METHODS

While cavitation prediction models can to some extent predict features such as cavitation flow fields, they cannot fundamentally alleviate the effects caused by cavitation. Therefore, corresponding cavitation suppression methods have been developed based on the study of cavitation instability mechanisms and cavitation prediction methods. How to effectively suppress cavitation damage is also an important research direction in the current cavitation research. Cavitation phenomena cause erosion and vibrations in hydraulic machinery. This research is mainly conducted with hydrofoils, propellers, etc., as the research subjects. Based on whether external energy input and consumption is needed, the suppression effects produced by these control technologies are examined from the perspectives of active cavitation suppression methods and passive cavitation suppression methods.

A. Active cavitation suppression methods

Active control methods refer to the use of external forces to control the structure of cavitation flow, thereby influencing the cavitation flow characteristics in the near-wall region,¹⁰⁹ and thus achieving the effect of suppressing cavitation. Its main form is to inject external energy into the flow field, such as water, air, polymers, and ultrasonic waves. Table III summarizes some active cavitation suppression methods.

In current active cavitation suppression research, the commonly used forms of injected energy are gas (air)^{46,110–113} or water.^{114–118} In 1995, Arndt *et al.*¹¹⁰ conducted experimental tests on a hydrofoil with a NACA0015 cross section. Small holes were set up at the leading edge of the hydrofoil, and air was injected. The method of air injection reduced pressure fluctuations, and under the condition where the normalized air injection rate was only 0.0004, the measured cavitation indexes significantly decreased, equivalent to a 2% cavitation rate. Reisman *et al.*¹¹¹ continuously injected air into oscillating hydrofoils with two different cross sections in a low turbulence water tunnel and ejected it from the suction side of the hydrofoil to study the impact of continuous air injection on the cavitation flow structure. Multiple sensors were arranged in the experiment to measure the unsteady pressure of cavitation on the hydrofoil surface. The study found that air

injection reduced cavitation noise by two orders of magnitude and possibly hindered the development of reentrant jets. The visualization during air injection is shown in Fig. 8.¹¹² From Fig. 8(b), attached cavitation, large-scale shedding cloud cavitation, etc., can be observed. The injected air developed downstream through the vortex core. Additionally, it was found that, under the same experimental conditions, pulsed air injection had a better suppression effect on cavitation noise. Ventilation affects pressure fluctuations, cavitation development, and turbulence structures. Injecting air can significantly improve the pressure fluctuations on the suction side of the hydrofoil, alter the development of cavitation, reduce the cavitation growth cycle, and decrease the turbulence structure scale in the wake flow field.⁴⁶ In addition to achieving cavitation suppression, the gas injection method is also applied in areas such as drag reduction for underwater vehicles.^{119–121}

In addition to the air injection method mentioned above, water injection can also be used to improve the cavitation flow characteristics near interfaces such as the hydrofoil surface, thereby suppressing cavitation. In recent years, research has mainly focused on placing jet holes on the surface of hydrofoils to weaken the development of cavitation. Lu *et al.*¹¹⁴ arranged micro-jet holes on the suction side of a NACA0066 hydrofoil, injecting water laterally into the hydrofoil. They used high-speed photography to study the cavitation characteristics on the hydrofoil surface and varied the jet velocity and flow rate to create different cavitation conditions for the hydrofoil. The study found that the arrangement of jet holes could completely suppress sheet cavitation. However, for cloud cavitation, active jets could weaken the growth and development of cavitation to a certain extent, but they could not completely avoid the phenomenon of cloud cavitation shedding. Additionally, as the cavitation number decreased, the effectiveness of jet holes in suppressing cavitation gradually weakened. Wang *et al.*¹¹⁵ adopted a similar approach, actively injecting water into a NACA66 (MOD) hydrofoil, with water continuously jetting vertically from the suction surface of the hydrofoil's chamber. By setting different jet velocities and arranging two jet hole positions, the effects of different factors on cavitation suppression were explored. Wang *et al.*¹²² injected water into a NACA66 hydrofoil to investigate the inhibition effect on cloud cavitation by adjusting the jet angle, jet position, and jet

TABLE III. Some active cavitation suppression methods.

Suppress objects	Injecting energy form	Cavitation suppression effect	References
NACA66 hydrofoil	Air	It accelerated the collapse of bubbles and reduced pressure fluctuations.	46
NACA0015 hydrofoil	Air	It increased the internal pressure of the bubble and significantly suppresses cavitation.	110
Oscillating hydrofoil	Air	It reduced cavitation noise and hindered the development of reentrant jet.	111
NACA0066 hydrofoil	Water	It completely suppressed sheet cavitation and weakens the development of cloud cavitation.	114
NACA66 (MOD) hydrofoil	Water	It suppressed the size of cavities.	115
NACA0012 elliptical hydrofoil	Water	It reduced the generation and intensity of blade tip vortex cavitation, affecting the initial structure of blade tip vortex cavitation.	116
Propeller	Water and polymer solutions	It increased the radius of the vortex core and delays the onset of tip vortex cavitation.	128
Venturi tube	Ultrasonic	The gas core was destroyed, and bubble cavitation was completely eliminated.	131

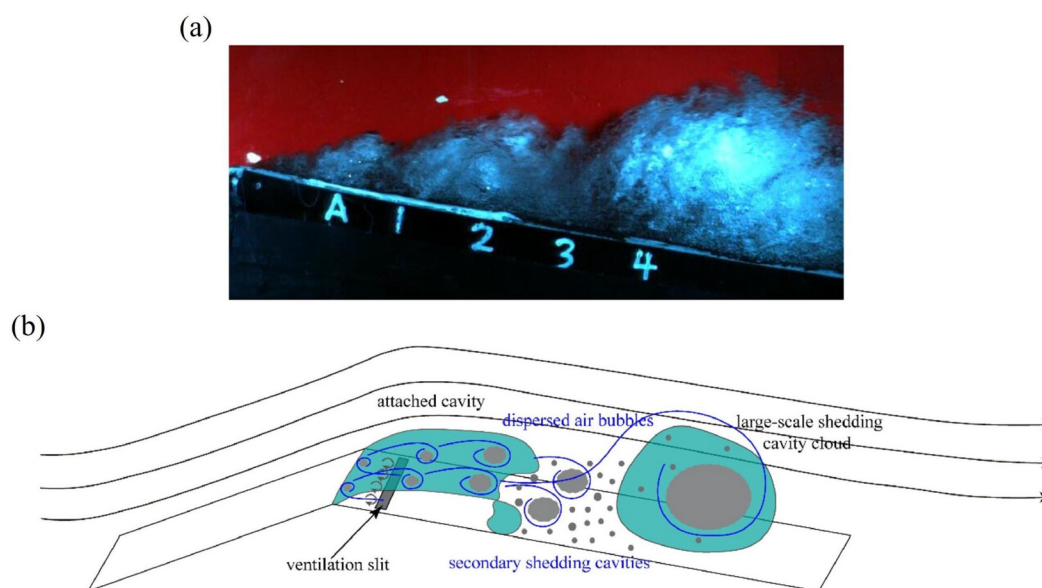


FIG. 8. The visualization during air injection: (a) Cavitation pattern diagram, and (b) sketch of cavitation structure. Reproduced with permission from Wang *et al.*, *Int. J. Multiphase Flow* **106**, 1–20 (2018). Copyright 2018 Elsevier Ltd.¹¹²

velocity. Active jets could effectively suppress the size of attached cavities, and different jet flow rates and jet positions could suppress cavitation development to varying degrees. Furthermore, tip leakage vortex cavitation can present risks to hydraulic equipment.^{123–125} Both tip jets and side jets had a significant cavitation suppression effect on tip vortex cavitation (TVC), with side jets directly affecting the original structure of TVC. Moreover, increasing the transverse velocity component of the jets could significantly enhance the cavitation suppression effect.¹¹⁶

In tip vortex cavitation, polymer injection is commonly used to suppress cavitation. This is achieved by creating small holes at the tips of propeller blades and injecting polymers through them. The presence of polymers reduces the possibility of cavitation occurrence, delays the onset of vortex cavitation, decreases the pressure drop at the vortex cavitation center, and can effectively alleviate tip vortex cavitation on propeller blades.^{126–128} Furthermore, studies have shown that ultrasonic pressure modulators (UPMs) can be utilized as an active control method to suppress cavitation.^{129–131} Dynamic sliding mode control can also be used to suppress the high frequency flutter of the signal by using an integrator and to force the bubble to decrease through proportional integral state feedback, thereby suppressing cavitation.¹³²

B. Passive cavitation suppression methods

Currently, most passive control methods have been developed based on the mechanism of cavitation instability. Passive control methods refer to those that do not rely on external power sources; instead, they often adjust the cavitation flow development near the wall region by altering the geometric structure or material properties of the hydraulic equipment surface itself. This optimization of cavitation flow structure aims to suppress cavitation phenomena effectively. Table IV summarizes several passive cavitation suppression methods, which mainly include the arrangement of vortex generators,^{133–136} surface

obstacles, etc. Figure 9 illustrates common passive cavitation control methods.

Positioning micro triangular prism vortex generators (micro-VGs) at the leading edge of hydrofoil [Fig. 9(a)] can alter the surrounding flow vectors. These micro-VGs generate counter-rotating vortices in their adjacent back regions, which change the pressure distribution and turbulent kinetic energy in the flow field, reduce pressure fluctuations, and further improve flow field stability.¹³⁷ Cylindrical vortex generators [Fig. 9(b)], also known as cylindrical cavitation bubble generators (CCGs), can influence the turbulent structures of unsteady flows at different cavitation stages. CCGs significantly reduce pressure fluctuations and can suppress a wide range of cloud cavitation. Moreover, during the cloud cavitation shedding stage, the key contributing factors such as axial and lateral jets to the formation of cloud cavities are effectively mitigated.¹³⁸ The formation and development of attached cavitation are influenced by the form of boundary layer separation. The thickness of the boundary layer induced by shockwaves is closely related to the height of the vortex generator.¹³⁹

Installing obstacles on the surface of hydrofoils has been proven to be an important passive control method for suppressing cavitation.^{140–143} Yin *et al.*¹⁴² placed obstacles at four different locations on the surface of a NACA0009 hydrofoil. They found that when the obstacles were placed at 30% of the chord length from the leading edge, the risk of cavitation was minimized compared to other positions. When obstacles were located at 10% or 20% of the chord length from the leading edge, their effect was similar to that of the previously mentioned vortex generators. The presence of obstacles increased the pressure in the near-wall region at the leading edge of the hydrofoil, effectively suppressing sheet cavitation. Under conditions of local cavitation oscillation, the reentrant jet propagating toward the leading edge would be impeded by the obstacles, leading to the transformation of large-scale cloud cavitation into smaller-scale cavities, thereby

TABLE IV. Some passive cavitation suppression methods.

Hydrofoil type	Suppression method	Position	Cavitation suppression effect	References
NACA0015	Micro-vortex generator	Hydrofoil leading edge	It suppressed the pressure fluctuations and the collapse of the cloud cavitation	137
CAV2003	Cylindrical cavitation bubble generators	Near the middle of the hydrofoil surface	It reduced the pressure pulsation at the trailing edge of the hydrofoil and alleviated cloud cavitation damage behavior	138
NACA0009	Obstacle	Four positions on the hydrofoil surface	It reduced the risk of cavitation and increased the lift to drag ratio.	142
NACA0015	Obstacle	Hydrofoil suction surface	It suppressed sheet cavitation and hindered the development of reentrant jet	143
NACA66	Obstacle	Trailing edge of hydrofoil	It weakened the formation of cloud cavitation and effectively controlled the reentrant jet.	144
NACA66	V-shaped groove	Hydrofoil suction surface	It significantly reduced cavitation at the trailing edge of the hydrofoil, and the pressure impact caused by cavitation collapse was significantly weakened.	150
NACA0009	Variable-depth groove	Side near the leading edge of the hydrofoil	It suppressed the tip leakage vortex.	151
NACA0015	Micro wedge structure	Near the hydrofoil leading edge	It reduced the turbulence energy intensity in the initial stage of cavitation and delayed the period of change in cavitation volume.	152
NACA0012	Microchannel	Near the hydrofoil leading edge	It suppressed the upper low-pressure area and reduced the scale of cavitation by nearly half.	153
NACA0015	Bionic fin-fin structure	Hydrofoil suction surface	It could reduce drag and eliminate the influence of lateral jet on the near wall surface	155
NACA0015	Bionic jet structure	Hydrofoil surface	It suppressed the shedding of large-scale vortex structure.	156
Hydrofoil section (representing turbine blades)	Leading-edge tubercles	Tip	It could reduce tip vortex.	158

preventing the shedding of large-scale cloud cavities. However, if the obstacles were placed too close to the leading edge of the hydrofoil, they could induce shear cavitation.¹⁴³ Additionally, recent studies have shown that placing obstacles at the trailing edge of the hydrofoil can also help to prevent the formation of cloud cavitation,¹⁴⁴ thereby reducing the impact of the enormous impact force generated by cloud cavitation collapse on the lifespan of hydraulic machinery. The height of obstacles can also have varying degrees of impact on cavitation suppression. Within a certain range of obstacle heights, the extent of cavitation suppression increases with the height of the obstacles.¹⁴⁵ Similar to obstacles, a porous material layer is added to the surface of the flow passage component, which affects the pressure around the porous layer and suppresses the cavitation in the primary stage.¹⁴⁶

In addition to vortex generators and surface obstacles, other common passive control methods such as grooves^{147–151} have garnered significant interest among researchers. Jia *et al.*¹⁵⁰ implemented a V-shaped groove design on the suction surface of a NACA66 hydrofoil. During the cavitation development stage, the groove inhibited the development of low-pressure and low-velocity regions on the hydrofoil surface, effectively suppressing cavitation at the leading edge of the hydrofoil. During the cavitation collapse stage, the presence of the groove reduced the cavitation volume fraction by 17.46% and extended the cavitation cycle. Additionally, the grooved hydrofoil reduced drag by 2.95% compared to the original hydrofoil while effectively suppressing cavitation. Han *et al.*¹⁵¹ studied a passive cavitation suppression method using variable-depth (VD) grooves on a NACA0009 hydrofoil

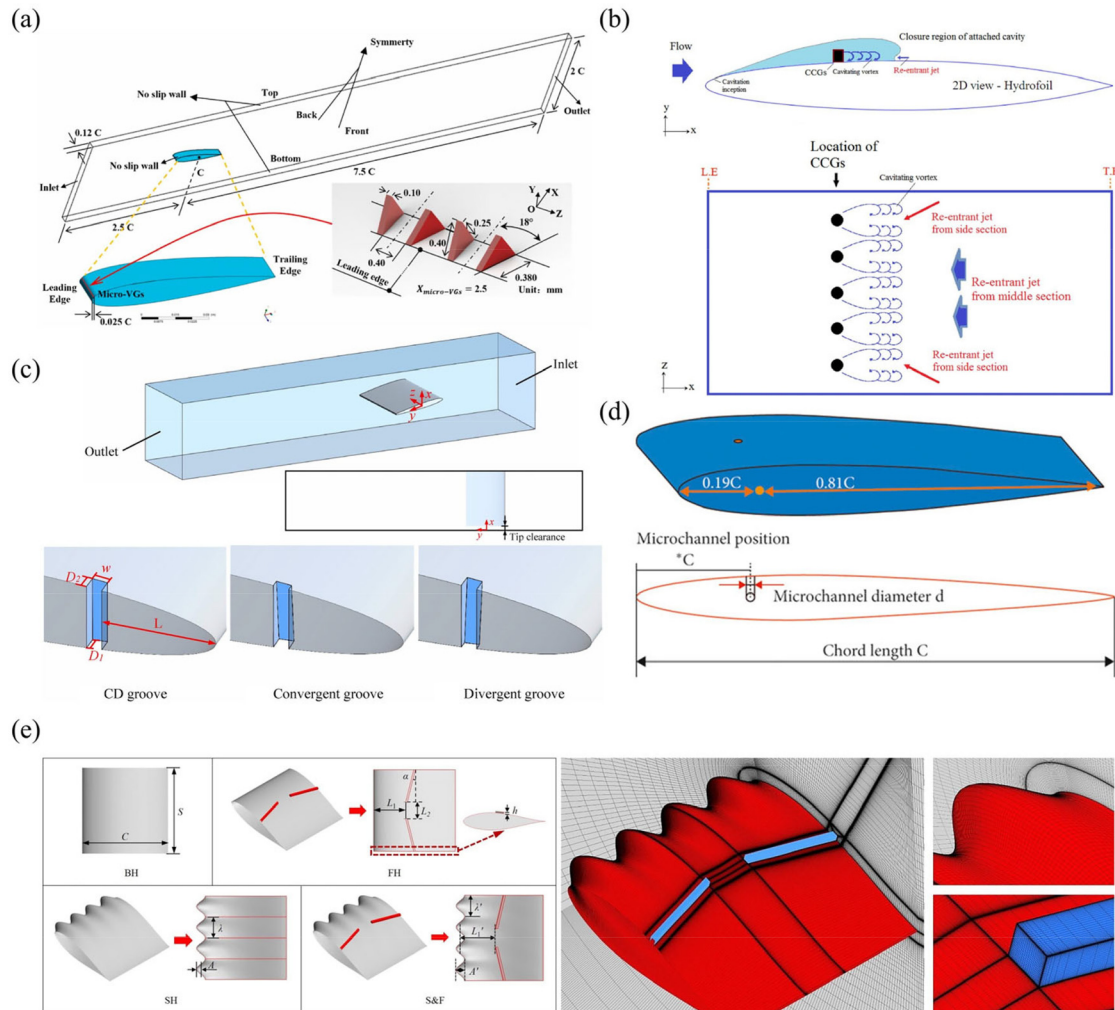


FIG. 9. Common passive cavitation control methods: (a) Placing vortex generators at the leading edge of the NACA0015 hydrofoil. Reproduced with permission from Qiu *et al.*, *Ocean Eng.* **297**, 117004 (2024). Copyright 2024 Elsevier Ltd.¹³⁷ (b) positioning cylindrical cavitation bubble generators near the midsection of the CAV2003 hydrofoil surface. Reproduced with permission from Kadirav *et al.*, *Int. J. Multiphase Flow* **125**, 103186 (2020). Copyright 2019 Elsevier Ltd.¹³⁸ (c) Installing variable-depth (VD) grooves near the leading edge on the sides of the NACA0009 hydrofoil. Reproduced with permission from Han *et al.*, *Renewable Energy* **199**, 546–559 (2022). Copyright 2022 Elsevier Ltd.¹⁵¹ (d) establishing microchannel on the NACA0012 hydrofoil. Reproduced with permission from Shi *et al.*, *Shock. Vib.*, **2021**, 6641839. Copyright 2021 Authors, licensed under a Creative Commons Attribution (CC BY) License.¹⁵³ and (e) designing a novel dual biomimetic structure on the NACA 634-021 hydrofoil. Reproduced with permission from Yang *et al.*, *Ultrason. Sonochem.* **102**, 106745 (2024). Copyright 2024 Elsevier Ltd.¹⁶⁰

to address tip leakage vortices (TLV) and tip leakage vortex cavitation (TLVC) [Fig. 9(c)]. The VD groove design increased the local pressure in the vortex core region, thereby reducing primary tip leakage vortex cavitation. The VD grooves included CD grooves, convergent grooves, and divergent grooves. All three types of grooves reduced primary TLV and secondary TLVC, with the divergent groove performing the best in terms of suppressing vortex cavitation. However, small-scale local vortices were generated near the grooves, and both convergent grooves and CD grooves led to significant vortex cavitation at the trailing edges of the grooves.

Inspired by flow separation control methods in aerodynamics, Qiu *et al.*¹⁵² proposed a passive cavitation suppression method based on micro wedge structures (Micro WS) on the surface of hydrofoils.

By placing Micro WS on the surface of a NACA0015 hydrofoil, they aimed to suppress attached cavitation on the hydrofoil surface and achieve turbulent flow, thereby preventing flow separation. The study found that the height of Micro WS determined by the boundary layer thickness is a critical parameter affecting cavitation performance. When the height of Micro WS was 0.05 mm and the chordwise position was 3.5 mm, the best cavitation suppression effect was observed, with a suppression rate of cavities reaching 16.7%. In recent years, a novel passive cavitation suppression technology called microchannel technology has emerged. Shi *et al.*¹⁵³ were the first to propose establishing microchannel on a NACA0012 hydrofoil that connect the internal and external parts of the hydrofoil [Fig. 9(d)]. By replenishing fluid in the internal low-pressure region, they aimed to suppress

cavitation. The cavitation suppression effect varied with different channel positions. Under certain microstructure conditions, larger channels could completely suppress cavitation, while relatively smaller ones could suppress cavitation by more than 50%. However, the design of microchannel could lead to the generation of unilateral vortices as the fluid flows through, exacerbating vortex cavitation. Additionally, if the pressure at the microchannel outlet exceeded an optimal value, the cavitation suppression effect would be weakened. Despite these challenges, microchannel technology represents an innovative passive control method that offers valuable insights and guidance for designing cavitation suppression methods in other hydraulic equipment.

In addition to the aforementioned passive cavitation suppression methods, some research is based on biomimetic principles and structures derived from marine organisms.¹⁵⁴ Placing a bionic fin-fin structure on the suction surface of a hydrofoil can effectively reduce turbulent kinetic energy, control the shedding process of vortex cavitation, and stabilize the flow field.¹⁵⁵ Inspired by the shark gill slit jet controlling the surrounding flow field, Gu *et al.*¹⁵⁶ placed bionic jet structures at various chordwise positions on a NACA0015 hydrofoil. They studied the cavitation flow characteristics on the hydrofoil surface using vortex identification methods. Compared to the original hydrofoil, when the jet was positioned 60 mm from the leading edge, the hydrofoil with the bionic jet structure significantly suppressed the shedding of large-scale vortex structures, achieving the best cavitation suppression effect. By studying the fish gill respiration, the bionic principle is used to convert the gill jet, and a periodic jet scheme is designed and added to the Clark-Y hydrofoil to weaken the cavitation development by affecting the velocity distribution and pressure gradient.¹⁵⁷ Additionally, the leading-edge tubercles can affect cavitation flow around the hydrofoil, thereby suppressing cavitation. Applying the biomimetic feature of leading-edge tubercles from humpback whale flippers to turbine blades can improve the lift-to-drag ratio of the hydrofoil and reduce tip vortices of the hydrofoil.¹⁵⁸ Additionally, coupling the jet angle with jet speed and performing numerical simulations on the cavitation characteristics of jet-equipped hydrofoil surfaces show that the cavitation suppression effect is most significant when the jet is ejected at a small angle and high speed along the hydrofoil surface.¹⁵⁹ Yang *et al.*¹⁶⁰ proposed a novel biomimetic structure [Fig. 9(e)], combining the leading-edge protuberances of the humpback flipper and the spine structure of the tail fin of the sailfish. Compared to a single biomimetic structure, this novel biomimetic structure not only reduced the total amount of cavitation by approximately 43% but also enhanced the stability of the flow field. In addition to marine life, bird feathers can provide technical inspiration. Installation of bionic raised stripes on NACA66 films inhibits the shedding of cavitation and reduces the size of cavitation bubbles.¹⁶¹

In summary, both active and passive control methods have their own distinct advantages and characteristics. Passive cavitation suppression methods do not require additional energy consumption, derive their acting force from the system itself, and typically have fixed control parameters, making them relatively simple to implement. On the other hand, active suppression methods are more suitable for applications in dynamically changing flow fields and offer better adaptability to complex cavitation flows. Under varying operating conditions, active methods can achieve excellent cavitation suppression effects by dynamically altering characteristic parameters and introducing

appropriate disturbances into the flow field, thereby achieving coupling with cavitation flows.

V. RESEARCH SUMMARY AND PROSPECTS

Through a summary of research on cavitation instability mechanisms, numerical simulations, cavitation prediction, and cavitation suppression methods, it is concluded that reentrant jets dominate cavitation flow instability. In addition, shock waves induced by cavity collapse also influence cavitation flows. With advancements in computational fluid dynamics (CFD) and computer hardware and software performance, numerical simulation has become a primary tool for studying cavitation phenomena. The selection of appropriate cavitation and turbulence models significantly impacts the accuracy of cavitation prediction and the understanding of cavitation instability mechanisms. Moreover, machine learning is playing an increasingly important role in this field. Beyond investigating the mechanisms and predicting cavitation, effectively suppressing the adverse effects of cavitation on hydraulic machinery is another key research direction. Active and passive control methods, such as injecting water or air, or modifying the surface structure or material properties of hydraulic equipment, are employed to mitigate cavitation. Although cavitation prediction methods for hydrofoils have demonstrated significant potential in predicting hydrodynamic performance and cavitation flow characteristics, different cavitation suppression methods have managed to reduce cavitation phenomena to a certain extent, there is still much work to be done in these areas that requires further study:

- (1) The cloud cavitation shedding mechanism caused by reentrant jets has already been widely studied and discussed, and most studies consider it the dominant factor in cavitation flow instability. Shockwaves, as one of the mechanisms for cloud cavitation shedding, also possess certain research value. Theoretically, the rupture of cavities can trigger shockwaves. However, issues such as the instantaneous nature of shockwaves lead to difficulties in adequately demonstrating their effects during actual cavitation processes on hydrofoil surfaces. Studying the role of shockwaves during cavitation flow processes poses certain challenges.
- (2) Although active and passive cavitation suppression methods each have their own unique advantages, they also have limitations. The balance and optimization between hydrodynamic performance and cavitation characteristics may become one of the main research focuses in the field of hydraulic machinery in the future.
- (3) Traditional numerical calculations often involve issues like high consumption of computing resources and high time costs. With the continuous development of computer hardware and software, artificial intelligence has experienced rapid growth. Applying artificial intelligence algorithms in hydraulic equipment has become a research trend, and sensible use can also save computational resources more effectively. How to choose the appropriate intelligent algorithm or model for different hydraulic equipment models, while accurately simulating cavitation flow field characteristics and the shapes of cavities at different stages of cavitation, is expected to become a hot research topic in the future.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Qianfeng Qiu: Data curation (equal); Visualization (equal); Writing – original draft (equal). **Yunqing Gu:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal). **Yun Ren:** Methodology (equal); Software (equal). **Chengqi Mou:** Investigation (equal); Methodology (equal). **Chaoxiang Hu:** Data curation (equal); Writing – original draft (equal). **Hongxin Ding:** Methodology (equal); Writing – original draft (equal). **Denghao Wu:** Software (equal); Writing – original draft (equal). **ZhenXing Wu:** Methodology (equal); Software (equal). **Jiegang Mou:** Conceptualization (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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